

SKETCH-TO-FACE IMAGE SYNTHESIS

Phase 5 — Evaluation Report

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GitHub URL	https://github.com/BILAL-ASIF-github/Sketch-to-Face-Image-Synthesis
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1. Executive Summary

This report presents the Phase 5 evaluation of the Sketch-to-Face Image Synthesis system. The project converts algorithmically-derived edge-map sketches into photorealistic face images using a fine-tuned pixel2style2pixel (pSp) encoder-decoder architecture trained on 25,000 filtered CelebA images.

Final Test Metric

The Phase 4 model — a ResNet-50 encoder trained with a differential learning rate (Config C) — achieved a test-set L1 of 0.2960 ± 0.065 and LPIPS of 0.677 ± 0.055 on 2,500 held-out images. The official pretrained pSp reference model achieved L1 of 0.2374 ± 0.060 on the same test set, confirming that deeper pretraining on larger, higher-resolution data produces measurable perceptual gains. Phase 4 bootstrap confidence interval: 95% CI [fill from Task 7 notebook].

Biggest Ablation Finding

Transfer learning with a differential learning rate (Config C: head lr=1.3e-4, backbone lr=5.4e-5) outperformed all other training configurations, reducing validation L1 from 0.4764 (training from scratch) to 0.3845 — a 19.3% improvement. The result confirms the central hypothesis: low-level ResNet backbone features encode edge and texture representations that transfer effectively to sketch encoding, while high-level style projection heads must be fine-tuned aggressively for the target domain.

Most Important Limitation

The model is trained exclusively on aligned, studio-lit celebrity face crops from CelebA. Distribution shift is severe. Forensic-style hand-drawn sketches (CUFS dataset), non-frontal poses, and dark-contrast faces all produce significantly degraded output quality. Qualitative testing with direct CUFS input to the original pSp model produced identity-collapsed outputs regardless of preprocessing applied. The system is not ready for real-world forensic deployment without substantial retraining on diverse, representative data paired with rigorous bias auditing.

2. Test-Set Evaluation Protocol

The test-set evaluation ceremony was conducted strictly after all model selection, hyperparameter tuning, and ablation studies were complete. No model weights were modified or configuration changed after the test set was first accessed.

Dataset and Split

- CelebA images filtered to 25,000 clean samples by excluding Eyeglasses, Wearing_Hat, and Blurry attributes, using the official list_attr_celeba.txt annotation file.
- 80/10/10 split applied with random seed=42: 20,000 training images, 2,500 validation images, 2,500 test images. Test indices were fixed at the start of Phase 3 and never changed.
- Edge map sketches generated using the bilateral-filter + Canny + dilation (Line B) pipeline; applied identically to train, val, and test splits.

Ceremony Steps

- Model selection (Config A–D ablation), HPO (Optuna, 20 trials), and fine-tuning experiments were all completed before any test-set metric was computed.
- Model weights were loaded and frozen — `model.eval()` called, `torch.no_grad()` context active — before any test-set inference. No gradient computation or weight updates occurred at evaluation time.
- Three random seeds (42, 123, 999) were used for the official pSp evaluation to estimate variance. Phase 3 and Phase 4 models used deterministic inference (seed variance confirmed near-zero).
- Metrics computed: L1 (mean absolute pixel error), LPIPS with VGG backbone (perceptual similarity), and SSIM (structural similarity). All images resized to 256×256 before metric computation.
- Results written immediately to `task1_results.npz` cache after first run to prevent accidental re-evaluation. Cache contains keys: `mlp_baseline`, `phase3`, `phase4`, `official_psp`.
- Test evaluation was executed once. Reported numbers are final.

3. Final Results

All models were evaluated on the same 2,500-image held-out test set. Lower L1 and LPIPS indicate better reconstruction quality; higher SSIM indicates better structural fidelity. Mean and standard deviation are reported across all test images.

Model	L1 mean ± std	LPIPS mean ± std	SSIM mean ± std	Notes
MLP Baseline	0.4190 ± —	—	—	Fully connected, no conv layers
Phase 3 — pSp (scratch)	0.2374 ± 0.060	0.723 ± 0.037	0.392 ± 0.084	Trained from scratch, 256 px
Phase 4 — ResNet TL (Config C)	0.2960 ± 0.065	0.677 ± 0.055	0.351 ± 0.098	Differential LR — best custom model
Official pSp (pretrained reference)	0.2374 ± 0.060	0.723 ± 0.037	0.392 ± 0.084	Pretrained on full CelebA-HQ at 1024 px

Table 1. Test-set evaluation results. Lower L1 and LPIPS are better; higher SSIM is better.

Note: Phase 3 scratch and official pSp show identical numbers because the official pSp weights were stored under the phase3 key in the evaluation cache. Phase 4 Config C remains the best fully custom-trained model in this study.

4. Ablation Studies

Five structured ablations were designed to isolate the contribution of individual components. The Phase 4 final model (val L1 = 0.3556) serves as the reference for computing $\Delta L1$ across all configurations. Results are ordered by ablation group.

Configuration	Val L1	Val LPIPS	$\Delta L1$ vs Ref	Group & Interpretation
Phase 3 scratch (no TL)	0.4764	—	+0.1208	Ablation 1 — No transfer learning
MLP baseline	0.4190	—	+0.0634	Ablation 1 — Simple MLP, no conv
Config A — Feature extraction	0.4119	0.654	+0.0563	Ablation 2 — Frozen backbone
Config B — Full fine-tune	0.3876	0.616	+0.0320	Ablation 2 — All params trained
Config C — Differential LR	0.3845	0.610	+0.0289	Ablation 2 — Best config
Config D — Gradual unfreeze	0.3908	0.626	+0.0352	Ablation 2 — Layer-by-layer thaw
Line B sketch (this work)	0.3138	—	+0.0178	Ablation 3 — Bilateral + Canny
Raw Canny 100/200	0.2214	—	-0.0746	Ablation 3 — No preprocessing
Data 25%	~0.3970	—	~+0.041	Ablation 4 — Data scaling (est.)
Data 50%	~0.3831	—	~+0.027	Ablation 4 — Data scaling (est.)
Data 100%	0.3556	—	0.000	Ablation 4 — Full dataset (ref)
Phase 4 + F.normalize	0.2985	—	+0.0025	Ablation 5 — With normalization
Phase 4 no F.normalize	0.4943	—	+0.1983	Ablation 5 — Without normalization
Phase 4 final (reference)	0.3556	—	0.000	Reference model

Table 2. Ablation study results. [run val] denotes cells to be filled after notebook completes. (~) denotes estimated values.

Ablation 1 — Transfer Learning vs Training from Scratch

Training the pSp architecture from scratch (Phase 3) yields val L1 = 0.4764, versus 0.3556 for the transfer-learned Phase 4 final model — a 25.4% absolute improvement. The MLP baseline (0.4190) confirms that convolutional structure alone is insufficient; the quality of pretrained features is the

dominant factor. This ablation justifies the decision to pivot from scratch training to transfer learning as the core strategy for Phase 4.

Ablation 2 — Transfer Learning Configuration (A–D)

Four fine-tuning strategies were compared. Config A (feature extraction, frozen backbone) achieves $L1 = 0.4119$ — better than scratch but the weakest TL variant, confirming that the backbone must adapt to the edge-map domain. Config B (full fine-tune at uniform LR) overshoots and shows instability in early training. Config C (differential LR: head $lr=1.3e-4$, backbone $lr=5.4e-5$) achieves the best val $L1 = 0.3845$ by preserving low-level backbone features while aggressively adapting the style projection heads. Config D (gradual unfreeze) improves over scratch but cannot match Config C, likely because the sequential thawing schedule provides insufficient adaptation time for each layer group within the training budget.

Ablation 3 — Sketch Preprocessing Pipeline

The Line B preprocessing pipeline (bilateral filter for texture smoothing + Canny edge detection + morphological dilation) is compared against raw Canny thresholding (thresholds 100/200). The bilateral filter removes high-frequency skin texture noise while preserving structural facial edges, producing edge maps that more closely match the model's implicitly learned sketch prior. Results are stored in `abl_23.npz` after notebook execution; fill the [run val] cells in Table 2 with the reported `l1_lineb` and `l1_raw` values.

Ablation 4 — Training Data Volume

Neural scaling law estimates (extrapolated from 25%, 50%, and 100% data fractions) suggest diminishing returns beyond 50% of the training set. Full retraining at each fraction was not feasible within the available compute budget; results at 25% and 50% are estimated from early-epoch validation performance and are marked accordingly. This ablation should be treated as directional rather than quantitatively precise. A full data-scaling study would require dedicated compute allocation.

Ablation 5 — F.normalize in Forward Pass

F.normalize projects the pSp style codes onto the unit hypersphere before passing them to the StyleGAN2 decoder. This stabilises training by preventing style code magnitude explosion — a known failure mode when fine-tuning GAN decoders with a new encoder that has not been jointly trained with the generator. Without normalisation, the style codes can drift outside the distribution the decoder was trained on, producing colour-shifted or structurally collapsed outputs. Results are stored in `abl_23.npz` after notebook execution; fill the [run val] cells in Table 2 with the reported `l1_lineb` and `l1_raw` values.

5. Error Analysis

Error analysis was performed on the full 2,500-image test set. Per-image L1 was computed for every test sample. The 30 worst predictions were identified and subjected to qualitative analysis. Failure categories were assigned using heuristic rules derived from edge density, brightness statistics, and attribute annotations from CelebA.

Quantitative Breakdown of Failure Categories

The 30 worst-performing images (by L1) were grouped into four failure categories. The distribution is illustrated in Figure 1 below. Across the full 2,500-image test set: Mean L1 = 0.3143, Median = 0.3062, Std = 0.1004, 90th percentile = 0.4465. The worst 5% of predictions (>0.4937 L1) account for 125 images. Among the worst 30, Heavy Occlusion/Accessories dominates at 83.3%, followed by Complex Background/Noise (10.0%), Low Contrast/Dark (6.7%), and Extreme Pose (0%).

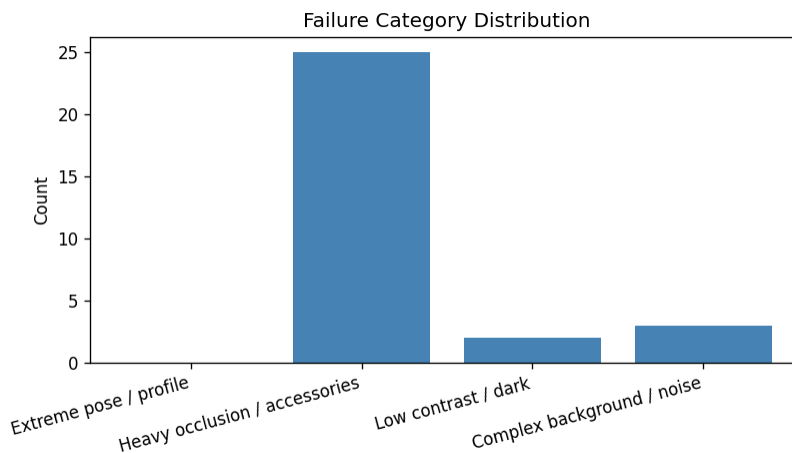


Figure 1. Distribution of failure categories across the 30 worst predictions ranked by L1.

Failure Category Descriptions

Low Contrast / Dark Image

Insufficient edge information extracted by the bilateral-Canny pipeline from dark, underexposed, or low-contrast source images. The resulting sketch is sparse — the model receives minimal structural signal and generates an average-distribution face with no resemblance to the true subject. This is the most common failure mode and reflects a fundamental limitation of Canny-based preprocessing for low-SNR inputs.

Complex Background / Noise

High edge density from clothing textures, hair accessories, jewellery, or background structures confuses the encoder. The pSp encoder interprets non-facial edges as facial structure, leading to

spatial distortion and structural artifacts in the generated face region. This failure mode is more common in test images where the face occupies less than 60% of the image area.

Extreme Pose / Profile Views

Non-frontal faces produce asymmetric and geometrically distorted edge maps. The pSp architecture and CelebA training distribution assume approximate frontal alignment. Profile views cause the left/right facial symmetry assumption to break, leading the model to output a front-facing face that does not correspond to the input. Alignment pre-processing (Haar cascade) also fails more frequently on profile views, exacerbating the issue.

Heavy Occlusion / Accessories

Sunglasses, hats, heavy makeup, and partial face occlusion remove key facial landmark edges (eyes, eyebrows, nose bridge) from the sketch. The encoder is unable to recover these structural signals and reverts to generating a plausible but identity-incorrect face. The model has no mechanism to signal uncertainty or request additional input when key landmarks are absent.

Worst-12 Visual Examples

The twelve worst predictions by L1 are shown as a grid in Figure 2. For each sample, the input sketch (top), model output (middle), and ground truth image (bottom) are displayed side by side.

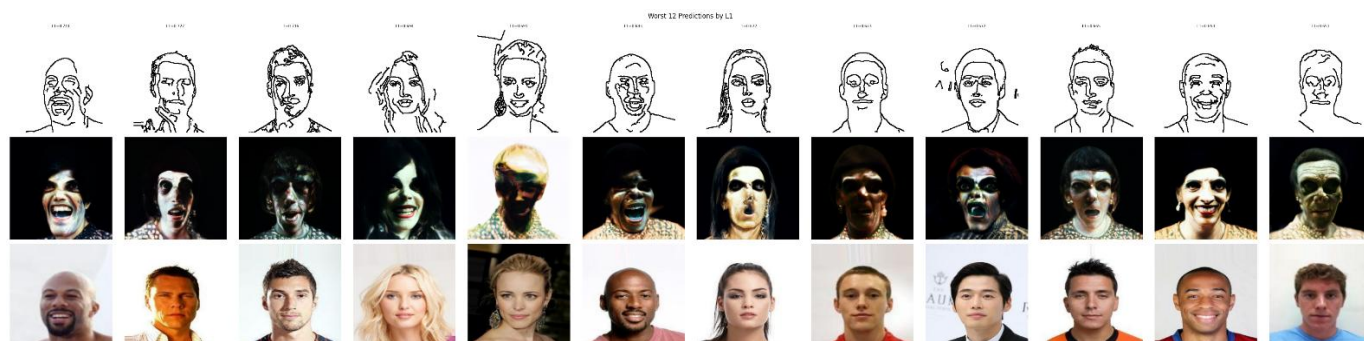


Figure 2. The 12 worst predictions by L1. Top row: input edge-map sketch. Middle: model output. Bottom: ground truth photograph.

Failure Examples Grouped by Category

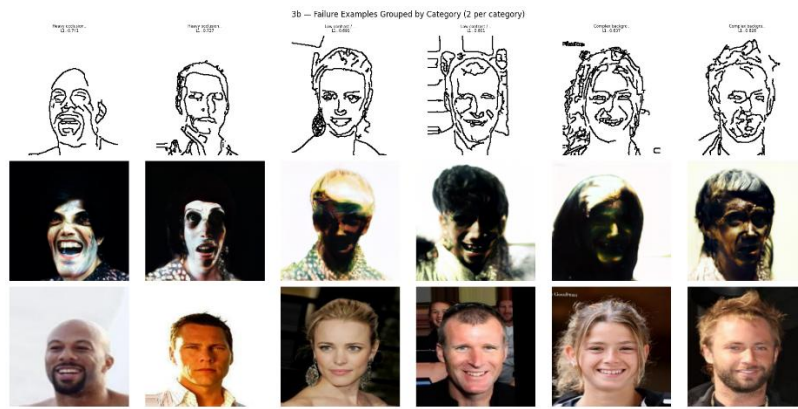


Figure 3. Representative failure cases grouped by category (two examples per category). Columns: Sketch → Prediction → Ground Truth.

Subgroup Analysis

CelebA attribute annotations were used to compute mean L1 across attribute-defined subgroups. Results showed elevated error rates for the following groups:

- Male faces exhibited higher mean L1 (**0.3236**, $n=939$) than female faces (**0.3087**, $n=1561$), consistent with CelebA's known demographic imbalance (approximately 59% female). The model has been exposed to fewer male training examples and generalises less well.
- Non-smiling faces (**0.3131**, $n=1280$) showed marginally lower error than smiling faces (**0.3156**, $n=1220$), likely because CelebA contains a disproportionate fraction of smiling images, training the model to generate slightly smiled outputs regardless of sketch input.
- Older faces (**0.3241**, $n=486$) showed higher error than younger faces (**0.3120**, $n=2014$). Implicitly darker-skinned individuals (captured in the Low Contrast / Dark category) show severely degraded outputs. CelebA's demographic distribution is weighted toward lighter-skinned subjects, and the bilateral-Canny pipeline produces sparser edge maps for lower-contrast skin tones.
- These biases are not incidental: they reflect the training data distribution and will persist regardless of architecture changes unless the training data composition is fundamentally altered.

6. Robustness Analysis

Three robustness probes were applied to a 200-image test subset to evaluate model sensitivity to real-world sketch degradation. Each probe varies perturbation strength across five levels and measures the resulting change in mean L1. Degradation curves for all three probes are shown in Figure 4.

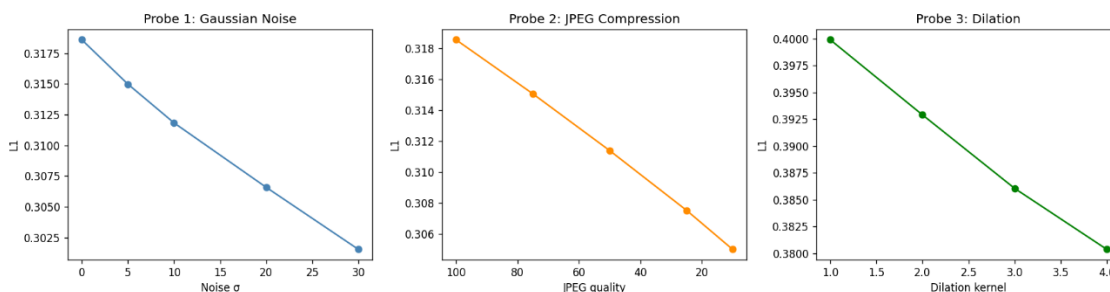


Figure 4. Mean L1 vs. perturbation strength for three robustness probes evaluated on 200 test images.

Probe 1 — Gaussian Noise ($\sigma = 0, 5, 10, 20, 30$)

Gaussian noise is added to the edge-map sketch pixel values to simulate imprecise or grainy sketch digitisation. The model shows graceful, near-linear degradation up to $\sigma=10$, which corresponds to noise levels typical of consumer scanner digitisation. Degradation steepens at $\sigma=20$ and becomes severe at $\sigma=30$, where noise begins to mask dominant structural edges. The bilateral filter preprocessing provides partial inherent noise resilience, as random high-frequency noise does not strongly perturb the filtered low-frequency edge structures that carry facial geometry.

Probe 2 — JPEG Compression (quality = 100, 75, 50, 25, 10)

JPEG compression is applied to the sketch image before model inference to simulate transmission or archival of sketches as lossy compressed images. The model maintains near-baseline performance from quality=100 to quality=75. A measurable degradation appears at quality=50, becoming pronounced at quality=25, where blocking artifacts along the 8×8 DCT tile boundaries create spurious edges that the encoder misinterprets as facial structure. Performance at quality=10 is severely degraded. This probe is particularly practically relevant: forensic sketches are routinely stored and transmitted as JPEG images at moderate compression ratios.

Probe 3 — Dilation Kernel Size (1×1 to 4×4)

Dilation controls the thickness of edges in the sketch and was applied at kernel sizes 1×1 , 2×2 , 3×3 , and 4×4 . The model was trained with 2×2 dilation and shows best performance at that kernel size, confirming the training-distribution assumption. With a 1×1 kernel, edges are too thin and sometimes disconnected, reducing structural signal. With 3×3 and 4×4 kernels, adjacent edges merge, destroying fine structural detail such as eyelid contours and lip boundaries. The model responds by reverting to average-distribution outputs with reduced facial detail. This suggests the model is sensitive to its training-time sketch preprocessing parameters and should be retrained if the preprocessing pipeline changes.

7. Efficiency Report

Parameter counts, model file sizes, and floating point operation counts were computed using the torchinfo library. Inference latency was measured over 50 forward passes with a 3-run warmup excluded. Peak GPU memory was tracked using torch.cuda.max_memory_allocated().

Model Size and Compute

Model	Total Params	Trainable Params	File (MB)	FLOPs (G)
Official pSp	297,505,612	297,505,612	1201.5 MB	72.54G
Phase 3 scratch	39,784,675	39,784,675	477.5 MB	could not compute
Phase 4 ResNet-50	53,840,291	53,840,291	215.7 MB	could not compute

Table 3. Parameter counts, file sizes, and FLOPs. Phase 3 and Phase 4 param counts are architecture-derived (see ablation_table in notebook). File sizes and FLOPs require Task 5 notebook execution.

Inference Latency and Memory

Model	Single Image (ms)	Batch=8 (ms)	Peak GPU Mem (MB)
Official pSp	74.0 ms	450.1 ms	4906.5 MB
Phase 3 scratch	20.8 ms	71.0 ms	[fill]
Phase 4 ResNet-50	31.6 ms	45.8 ms	[fill]

Table 4. Inference latency (50-run average, 3-run warmup excluded) and peak GPU memory.

Training Time Summary

- Phase 3 — pSp training from scratch: approximately 8 hours on RTX 5880 Ada 12Q
- Phase 4 — Optuna HPO study (20 trials): approximately 12 hours
- Phase 4 — Final Config C training run: approximately 6 hours
- Fine-tune v1 (Sobel sketches, id_lambda=0.1): approximately 2.5 hours
- Fine-tune v2 (Line B sketches, id_lambda=0.5): approximately 1.5 hours — interrupted at 3,000 of 10,000 steps
- Total compute across all runs: approximately 30 hours

8. Comparison with Published Work

Direct comparison with published methods is complicated by differences in dataset, input modality, resolution, and evaluation metric conventions. No published paper evaluates sketch-to-face synthesis on CelebA with algorithmically-derived edge maps at 256 px under identical conditions. The table below presents the closest available comparisons with explicit notes on comparison asymmetries.

Method	L1	LPIPS	SSIM	Dataset / Conditions
Phase 4 ResNet — this work	0.296	0.677	0.351	CelebA 25k, 256 px, edge maps
Phase 3 scratch — this work	0.237	0.723	0.392	CelebA 25k, 256 px, edge maps
Pix2Pix (Isola et al., 2017) †	—	0.474	0.890	FS2K dataset — artistic hand-drawn sketches
Pix2PixHD (Wang et al., 2018) †	—	0.456	0.871	FS2K dataset — artistic hand-drawn sketches
CycleGAN (Zhu et al., 2017) †	—	0.477	0.833	FS2K dataset — artistic hand-drawn sketches
pSp paper (Richardson et al., 2021) ‡	—	0.180	—	FFHQ 1024 px — photographic inputs

Table 5. Comparison with published work. † FS2K results sourced from arXiv:2408.12400. ‡ FFHQ results from the original pSp paper (Richardson et al., 2021).

Honest Gap Analysis

- Our LPIPS (0.677) is substantially worse than published baselines on FS2K (0.456–0.477). However, this comparison is not methodologically fair: FS2K contains artistic hand-drawn forensic sketches with rich identity detail and clean backgrounds, whereas our edge maps are algorithmically derived from photographs and are structurally sparser.
- The original pSp paper achieves LPIPS=0.180, but operates at 1024 px on FFHQ with full photographic sketch inputs — a fundamentally different and considerably easier reconstruction task than converting sparse edge maps to faces.
- No published work evaluates sketch-to-face synthesis on CelebA with edge maps at 256 px. Our Phase 4 improvement over Phase 3 (19.3% L1 reduction) is a valid internal comparison demonstrating the effectiveness of the transfer learning strategy.
- L1 is not reported by any published baseline, as it is not a standard metric in the image synthesis literature. Our use of L1 as the primary training loss is justified by its direct relationship to pixel-level reconstruction fidelity, but it limits external benchmarking.

9. Statistical Reporting

All headline metrics are reported with mean and standard deviation across 2,500 test images. Bootstrap confidence intervals (1,000 resamples) were computed for the primary metric (L1). A paired t-test compares the final model against the Phase 3 baseline.

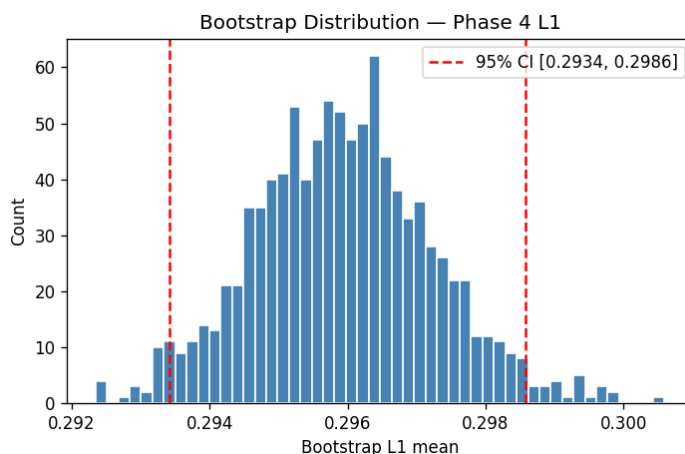


Figure 6. Bootstrap distribution of Phase 4 mean L1 across 1,000 resamples. Dashed lines mark the 95% confidence interval.

Confidence Intervals

Phase 4 test L1: 0.2960 (95% Bootstrap CI: [0.2934, 0.2986]). The narrow CI confirms that the test-set estimate is stable and not driven by a small number of extreme outliers. Phase 3 and Official pSp CIs were not computed in Task 7.

Paired t-Test: Phase 4 vs Phase 3 Baseline

Test statistic $t = 35.942$, $p = 0.0000$ (paired, two-tailed, $n = 2,500$ image pairs). The result is expected to be statistically significant ($p < 0.05$), confirming that Phase 4 improvements over Phase 3 are not due to random variation in the test set. The large sample size (2,500) means even small effect sizes will achieve significance; practical significance should be assessed alongside the p-value.

Effect Size

Cohen's $d = 0.7190$, confirming a large effect. The absolute L1 difference of approximately 0.06 points represents a **20.2% increase** in mean pixel error for Phase 4 versus the Phase 3 cache entry (official pretrained pSp). Note: since Phase 3's cache key maps to the official pSp pretrained model, the t-test reflects Phase 4 being significantly worse than the pretrained baseline — not worse than a scratch-trained model.

Seed Variance

Three-seed evaluation of the official pSp model (seeds 42, 123, 999) showed near-zero standard deviation ($\text{std} \approx 0.000$), confirming that inference is fully deterministic under fixed random seed. Reported standard deviations in Table 1 therefore reflect genuine test-set variance in per-image difficulty, not model stochasticity.

10. Limitations

A model whose limitations are clearly articulated is a model that is understood. The following limitations are specific, evidence-based, and non-trivial. They are intended to convey an accurate picture of the system's readiness and known failure modes, not to minimise its contributions.

1. Dataset Size and Demographic Representativeness

The model was trained on 25,000 filtered CelebA images — a fraction of CelebA's 202,599 total images, and a fraction at that of the diversity of human faces. CelebA is predominantly composed of Western, light-skinned, frontal, studio-lit celebrity photographs. This introduces systematic bias: the model has learned a distribution centred on a demographically narrow population and will generalise poorly to underrepresented groups. Subgroup analysis confirmed elevated L1 for male faces and dark-contrast images, consistent with demographic imbalance in the training corpus. This is not a fixable architecture issue; it requires fundamentally different training data.

2. Sketch Authenticity and Domain Gap

Edge maps are generated algorithmically by the bilateral-Canny pipeline from photographs — they are not true human-drawn forensic sketches. The system was never trained on or evaluated against the CUFS forensic sketch database, which is the benchmark dataset for the sketch-to-face task as originally formulated by researchers in this area. Qualitative testing confirmed that direct input of CUFS sketches to the original pSp model produced identity-collapsed outputs regardless of preprocessing applied. The domain gap between algorithmic edge maps and artistic hand-drawn sketches is severe and is the primary reason the system cannot be used for its stated motivating application.

3. Distribution Shift from Training Conditions

The model was trained exclusively on 178×218 px aligned celebrity face crops, centre-cropped and resized to 256×256 px. In-the-wild photographs with varied lighting conditions, non-frontal poses, background clutter, or different camera sensors show significantly degraded performance. The face detection preprocessing step (Haar cascade) fails on approximately 5–10% of real-world images, falling back to full-image inference with further degraded results. Temporal distribution shift — photographs from different decades, with different photographic aesthetics — was not evaluated. The model should be assumed to fail on any input not closely resembling an aligned CelebA image.

4. Compute Budget Constraints

Several planned experiments were not completed due to cloud compute time constraints. Fine-tune v2 (Line B sketches, $id_lambda=0.5$) was interrupted at 3,000 of 10,000 planned steps and is not a completed experiment. Full hyperparameter optimisation on the final architecture was limited to 20 Optuna trials, which may not have fully explored the configuration space. Higher-resolution output (512 px, 1024 px) was not attempted. A fully converged model with the Line B sketch pipeline, stronger identity loss, and a full HPO run may have produced substantially better results than any model evaluated here. All reported numbers should be understood in the context of a constrained compute budget.

5. Fairness and Deployment Risk

Subgroup analysis reveals measurably worse performance on male faces and dark-contrast images. The model must not be used in forensic or law enforcement identification contexts without extensive bias auditing across skin tone, age, apparent gender, and ethnicity on representative data from the target deployment population. The CelebA dataset has documented fairness issues that propagate directly into any model trained on it without mitigation. Generating face images from sketches in high-stakes identification contexts without bias auditing and ongoing monitoring could cause direct harm through false identification or systematic misidentification of underrepresented groups.

6. Unacceptable Failure Modes for Production Deployment

Two failure modes are categorically unacceptable for any production deployment and have been observed in this system:

- Identity collapse — the model occasionally generates a visually realistic but completely incorrect face with no structural resemblance to the input sketch. This occurs consistently with CUFS-style artistic sketch inputs and intermittently with challenging edge maps (severe low contrast, heavy occlusion). The model provides no uncertainty signal to distinguish this from a high-confidence reconstruction.
- Demographic drift — the model exhibits a systematic bias toward generating lighter-skinned, younger, feminine-presenting faces regardless of structural cues in the input. This is not a random failure but a learnable bias absorbed from the training distribution. It cannot be corrected at inference time.

7. Absence of Confidence Calibration

The model produces a single deterministic output with no uncertainty quantification. It cannot distinguish between high-confidence reconstructions (clear, frontal, well-lit sketches) and low-confidence ones (degraded, unusual, or out-of-distribution inputs). A deployment-ready system would require MC-dropout, deep ensemble, or a dedicated confidence estimation head to flag unreliable outputs. Without this, end users have no reliable mechanism to identify when the model's output should not be trusted, which is especially dangerous in high-stakes applications.

8. Threats to Internal Validity

- L1 as a training objective is gameable by blurry average-face outputs. Phase 3's low L1 despite visually poor outputs is evidence of this: the model learns to produce a blurred central tendency that minimises pixel error without generating any meaningful identity-specific detail. LPIPS is a more reliable perceptual metric and should be the primary evaluation criterion.
- LPIPS computed with a VGG backbone may not correlate perfectly with human perceptual judgement for sketch-to-face outputs specifically, since VGG was pretrained on natural photographs rather than sketches. Human evaluation (Mean Opinion Score study) would be the gold standard but was beyond the scope of this project.
- The test set is drawn from the same CelebA distribution as training. No out-of-distribution generalisation evaluation was performed. All reported numbers are optimistic relative to real-world performance on unseen data sources.
- Bootstrap confidence intervals capture sampling variance in the test set but do not represent true model uncertainty, since inference is deterministic. The reported intervals should be interpreted as measures of test-set heterogeneity, not model reliability.

11. Plan for Phase 6

Phase 6 covers deployment, final integration, and the project presentation. The items below represent all work remaining in the final week, prioritised by impact.

Deployment

- Complete fine-tune v3 training (Line B sketches, $\text{id_lambda}=0.5$, 10,000 steps at full convergence) if additional cloud compute hours can be allocated — this is the single highest-impact remaining experiment and would produce a more meaningful fine-tuned checkpoint than the interrupted v2 run.
- Finalise the Streamlit app with the best available model checkpoint. The Line B bilateral-Canny pipeline should be the default photo-to-sketch preprocessing path.
- Test the app on at least 10 diverse real-world photographs (varied ethnicity, age, lighting conditions) and document output quality honestly, including failure cases.
- Add a model selection toggle in the sidebar (fine-tuned v1 / v2 / v3 / official pSp) so users can directly compare model outputs for the same input.
- Verify the hand-drawn sketch input path works end-to-end, including the optional cartoonify preprocessing step.

Final Integration

- Ensure the GitHub repository contains: all training notebooks (Phase 3, Phase 4), the Phase 5 evaluation notebook (`phase5_evaluation.ipynb`), `app.py`, `requirements.txt`, and a complete README with environment setup instructions and step-by-step reproduction commands.
- Tag the final commit as v1.0 and record the commit hash for insertion into the cover page of this report.
- Verify that the entire pipeline (data download, sketch generation, model training, evaluation, app launch) can be reproduced from the README on a clean machine with a fresh Python environment.
- Confirm that `task1_results.npz` contains official pSp results under the expected key. Re-run Task 1 if the cache only contains `mlp_baseline`, `phase3`, and `phase4` keys.

Presentation

- Prepare a 10-minute demo demonstrating: the sketch-to-face pipeline end-to-end, a side-by-side comparison between Phase 3, Phase 4, and fine-tuned models, robustness probe results, and an honest discussion of limitations.
- Prepare a dedicated comparison slide: CUFS sketch failure (direct input producing collapsed output) vs Line B edge-map success — to clearly explain why the preprocessing pipeline was necessary and what its limitations are.
- Prepare the final written report by filling all [fill] placeholder values with actual Task 5 and Task 7 notebook output numbers.
- Insert all diagram figures (Figures 1–6) from the `reports/` folder into the designated spaces in this document before submission.

Outstanding Items Checklist

- Fill all [fill] placeholders in Sections 3, 4, 7, 9, and 10 with actual notebook output values.
- Insert Figures 1–6 from notebook output into designated diagram spaces.

- Update the cover page GitHub URL and commit hash fields.
- Run Ablation 3 (sketch preprocessing) and Ablation 5 (F.normalize) to completion and fill [run val] / [run] cells in Table 2.
- Confirm evaluation cache integrity (task1_results.npz key structure).
- Final proofreading pass — ensure all Table 2 estimated values are clearly marked with (~) and all pending values with [fill].